Homework #8. (Due April 20, 2025)

1. **Very small semiconductor crystals, composed of approximately 1000 to 10,000 atoms, are called quantum dots. Quantum dots made of the semiconductor CdSe are now being used in electronic reader and tablet displays because they emit light efficiently and in multiple colors, depending on dot size. The density of CdSe is 5.82*g*/*cm*3. What is the mass of one 6.5-nm CdSe quantum dot?**

2. A quantum dot has a radius of approximately 5nm. If the dot is made of silicon (Si), the mass of the dot is 1.2 × 10 −21 kg. How many total electrons are in the dot? The atomic mass of silicon is 28g/mole and the atomic number is 14. Don't forget to convert grams to kilograms

3. Orange CdSe quantum dots have a diameter of 7×10^{-7} cm. (That would be 7 nm, hence the term "nanoparticle.") Assuming quantum dots to be spherical, calculate the volume of an orange CdSe quantum dot in cm^3.

4. A blue CdSe quantum dot has a volume of 4×10^{-21} cm^3. Assuming the density of a CdSe quantum dot is the same as that of the bulk compound (4.82 g/cm^3), determine the mass of a blue CdSe quantum dot.

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On average, how many CdSe formula units would be contained in a blue CdSe quantum dot.